

AI Trainers – Technical Agent Creation Assignment

Focus: Practical AI building skills + domain adaptation

Assignment Overview

Candidates must **design and build ONE AI agent**, but architect it to be **adaptable across one of these 5 industries**:

- Pharma
- Medical
- Finance
- Marketing
- Hospitality

Core Requirement

Build a **specific-domain Agent** with:

- LLM-based reasoning (GPT / Azure OpenAI)
- Retrieval-Augmented Generation (RAG)
- Prompt engineering
- Basic guardrails (safety, accuracy)

Task Details

Step 1: Base Agent Design

Create a **general-purpose AI agent** that:

- Accepts user queries
- Uses:
 - Knowledge base (documents or dataset)
 - Contextual reasoning
- Produces:
 - Structured output (recommendations / summaries)

Step 2: Domain Adaptation (select a sector)

Candidate should show how the SAME agent adapts:

Pharma Agent

- Use case: Drug research assistant
- Tasks:
 - Summarize research papers
 - Highlight risks / interactions

Medical Agent

- Use case: Clinical decision support (non-diagnostic)
- Tasks:
 - Suggest possible guidelines
 - Summarize patient case notes

Hospitality Agent

- Use case: Guest experience optimizer
- Tasks:
 - Analyze reviews
 - Suggest improvements

Finance Agent

- Use case: Financial insights assistant
- Tasks:
 - Analyze financial reports
 - Generate investment insights (non-advisory)

Marketing Agent

- Use case: Campaign strategist
- Tasks:
 - Generate campaign ideas
 - Analyze customer segments

Step 3: Technical Requirements

Candidates must include:

1. **Architecture Design**
 - Components:
 - LLM
 - Vector DB
 - APIs (optional)
 - Diagram (simple is OK)
2. **Prompt Engineering**
 - System prompts for each domain
 - Few-shot examples
3. **RAG Implementation**
 - Sample knowledge base
 - Retrieval logic
4. **Evaluation Approach**
 - How accuracy is tested
 - Handling hallucination
5. **Guardrails**
 - Avoid medical/financial risk outputs
 - Confidence scoring or disclaimers

Step 4: Deliverables

- Code (LangChain / or another agent builders)
- Demo (live or recorded)
- Short document explaining:
 - Design decisions
 - Trade-offs
 - Scaling approach

Bonus Challenges (for top candidates)

- Multi-agent collaboration (e.g., Finance + Marketing insights)
- Memory-based personalization
- Integration with UI (chat interface)
- Use of tools (search, calculator, APIs)

Scoring Framework (Optional)

Area	Weight
Technical depth	30%
Problem-solving	25%
Real-world applicability	20%
Communication	15%
Innovation	10%

Good Luck .